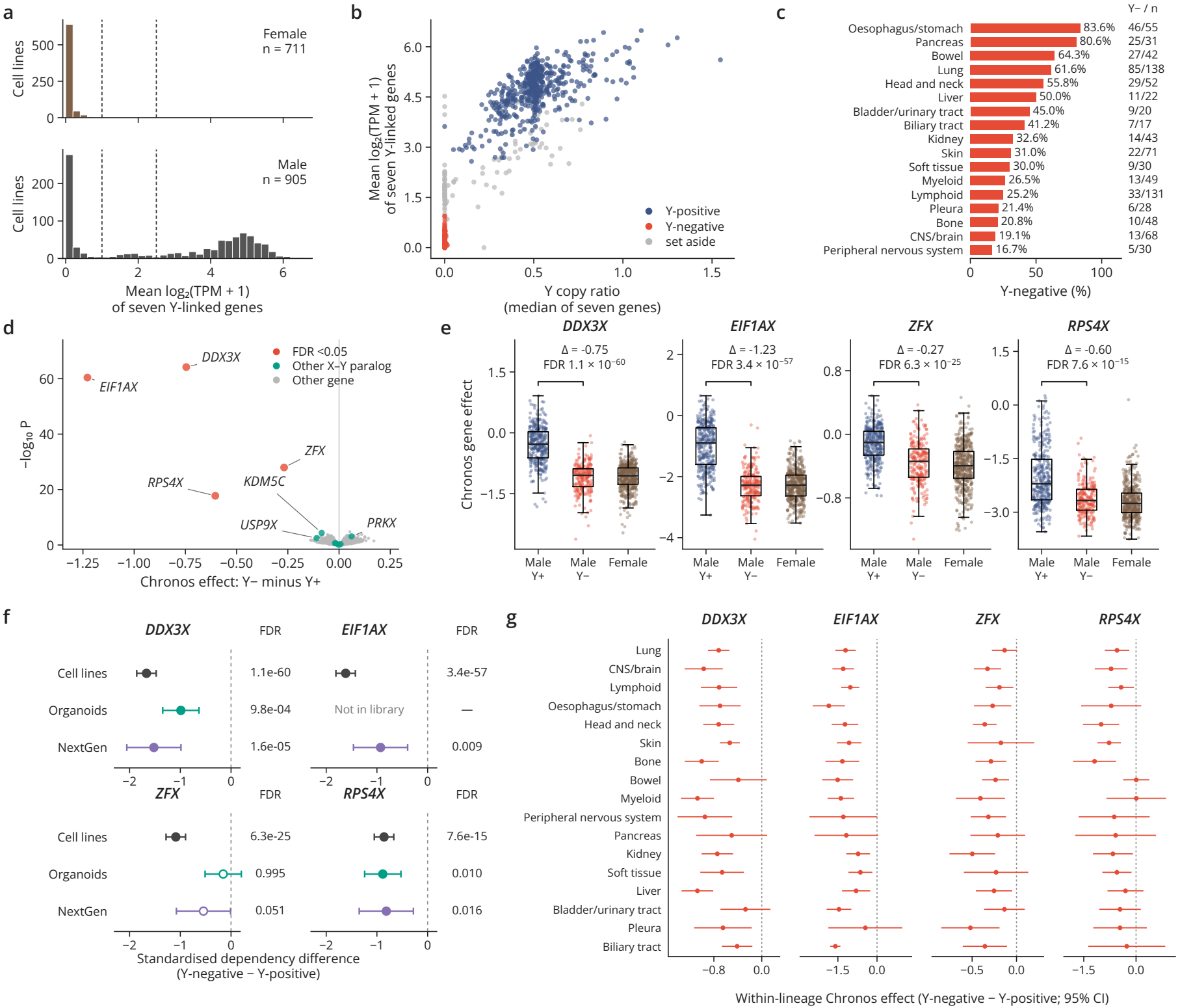

a
Female
n = 711
Male
n = 905
Cell lines
Mean log2(TPM + 1)
of seven Y-linked genes
b
Mean log2(TPM + 1)
of seven Y-linked genes
Y-positive
Y-negative
set aside
Y copy ratio
(median of seven genes)
c
Y− / n
Oesophagus/stomach 83.6% 46/55
Pancreas 80.6% 25/31
Bowel 64.3% 27/42
Lung 61.6% 85/138
Head and neck 55.8% 29/52
Liver 50.0% 11/22
Bladder/urinary tract 45.0% 9/20
Biliary tract 41.2% 7/17
Kidney 32.6% 14/43
Skin 31.0% 22/71
Soft tissue 30.0% 9/30
Myeloid 26.5% 13/49
Lymphoid 25.2% 33/131
Pleura 21.4% 6/28
Bone 20.8% 10/48
CNS/brain 19.1% 13/68
Peripheral nervous system 16.7% 5/30
Y-negative (%)
d
FDR <0.05
Other X–Y paralog
Other gene
DDX3X
EIF1AX
ZFX
KDM5C
RPS4X
USP9X
PRKX
−log10 P
Chronos effect: Y− minus Y+
e
DDX3X
Δ = -0.75
FDR 1.1 × 10−60
EIF1AX
Δ = -1.23
FDR 3.4 × 10−57
ZFX
Δ = -0.27
FDR 6.3 × 10−25
RPS4X
Δ = -0.60
FDR 7.6 × 10−15
Chronos gene effect
Male Y+
Male Y−
Female
f
DDX3X FDR EIF1AX FDR
Cell lines 1.1e-60 3.4e-57
Organoids 9.8e-04 Not in library —
NextGen 1.6e-05 0.009
ZFX FDR RPS4X FDR
Cell lines 6.3e-25 7.6e-15
Organoids 0.995 0.010
NextGen 0.051 0.016
Standardised dependency difference
(Y-negative − Y-positive)
g
DDX3X
EIF1AX
ZFX
RPS4X
Lung
CNS/brain
Lymphoid
Oesophagus/stomach
Head and neck
Skin
Bone
Bowel
Myeloid
Peripheral nervous system
Pancreas
Kidney
Soft tissue
Liver
Bladder/urinary tract
Pleura
Biliary tract
Within-lineage Chronos effect (Y-negative − Y-positive; 95% CI)